Fekete collocation points for triangular spectral elements

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Abstract. We propose a new spectral element method based on Fekete points. We use the Fekete criterion to compute points which are almost optimal for approximation, differentiation and quadrature. These grids have the same number of points as the dimension of the associated polynomial space, thus allowing us to use a cardinal function basis which leads to a diagonal mass matrix. For quadrilaterals, Gauss-Lobatto points are the unique Fekete points, making the method equivalent to the standard spectral element method. But unlike the Gauss-Lobatto points, Fekete points generalize to other domains such as the triangle. Furthermore, numerical and theoretical evidence suggests that the element boundary points are the Gauss-Lobatto points, making Fekete point triangular elements and quadrilateral elements naturally conform. Thus triangles and quadrilaterals can be combined in the same grid while retaining a diagonal mass matrix. We present an algorithm to compute Fekete points along with results for the triangle up to degree 19. For degree $d > 10$ these points have the smallest Lebesgue constant currently known.

Key words. triangle, Lebesgue constant, spectral elements, h-p finite elements, Fekete points

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1. Introduction. The spectral finite element method [22] is a spectrally accurate algorithm for solving differential equations on unstructured grids. Typically the computational domain is broken up into quadrilateral elements. Within each of these elements all variables are approximated by high degree polynomial expansions. The discrete equations are then derived using an integral form of the equations to be solved. When used with conforming elements and a clever choice of test functions and collocation points, the resulting mass matrix is diagonal [21]. This property is particularly useful for solving initial value problems like the shallow water and primitive equations used in geophysical modeling. For these equations, a diagonal mass matrix spectral element discretization will lead to a fully explicit method. This simple form of the method is perhaps the most natural way to achieve high order finite element methods. It is currently used in the global three-dimensional ocean and atmospheric models SEOM and SEAM [14,17,28]. Its exponential convergence under increasing polynomial degree has been shown for realistic nonlinear problems with many elements and degree as high as 56 [29].

At present, the diagonal mass matrix spectral element method is only available with conforming quadrilateral grids. This is because the method relies on the existence of high order quadrature formulas which use the same number of collocation points as basis functions. For the square, these methods use a tensor product of Gauss-Lobatto collocation points [10, p 57]. The quadrature formula associated with an $n \times n$ grid of these points will actually integrate exactly a polynomial space of dimension $(2n-2)^2$. This leads us to loosely define the term *Gauss-like quadrature* for two dimensional domains (like the triangle and the square) as a quadrature formula on N points which integrates exactly a functional space of dimension close to $4N$. An $n \times n$ grid of Gaussian points achieves this upper bound, integrating exactly a polynomial space of dimension $4n^2$.

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One drawback of this spectral element method is the requirement of conforming quadrilateral grids. These grids are quite complicated to generate. There are now automated programs to perform this task, such as [25], but the grids are rarely as uniform as triangulations. More uniform grids can be constructed if one adopts a non-conforming, mortar element approach like that taken in [4, 15], but then we loose the simple, fully explicit spectral element formulation.

The most natural alternative to this grid generation problem would be to use a triangular spectral element method. To generalize the quadrilateral element approach, we would need a Gauss-like quadrature formula for the triangle. But before we can attempt to construct such a formula, we need to choose the finite dimensional functional space we are interested in integrating. The space used in quadrilateral methods is given by the span of $\{x^m y^n,\ m,n \leq d\}$, which we call a *diamond truncation* because it has that shape when displayed using Pascal's triangle of the monomials in x and y (see [3, p. 157]). For triangles, the usual choice is a *triangular truncation* of polynomials, spanned by the monomials $\{x^m y^n,\ m+n \leq d\}$. We could also consider non-polynomial spaces. For example, to approximate periodic functions within our elements the best choice would be the trigonometric polynomials with a trapezoidal quadrature formula, otherwise known as the FFT. However, results from [31] (which we briefly describe in Section 4) confirm that the conventional triangular polynomial truncation is an appropriate choice for the triangle.

The most popular approach for triangular spectral element methods has been that of [11,24, 30]. These methods use the triangular polynomial truncation and the warped tensor product quadrature formula. Unfortunately, this quadrature formula is over-sampled: it has more grid points than we have basis functions, making it impossible to construct basis functions which lead to a diagonal mass matrix. Nor does this approach have a particularly easy-to-invert mass matrix. To develop a triangular spectral element method with a diagonal mass matrix, we need to find an accurate set of collocation/quadrature points for the triangle where the number of points is the same as the number of basis functions. Constructing Gauss-like quadrature formulas for the triangle turns out to be an extremely hard problem. There are a few, low-degree special cases, but there are also degrees for which it is known that Gaussian quadrature formulas do not exist. Many of these results for the triangle are summarized in [20]. Gaussian (and Gauss-Lobatto) type quadrature formulas seem to be extremely rare, and restricted to a few domains like the interval and its tensor products.

In this work we propose a new spectral element method for both triangles and quadrilaterals which retains the diagonal mass matrix. Instead of being based on Gauss-Lobatto quadrature points, it will be based on *Fekete points* [13]. Other references for Fekete points can be found in the recent paper [5], which summarizes many of the known results and gives some open questions. The references [6,23,1] represent the only work we are aware of in which Fekete points are computed in more than one dimension. A similar approach, only with collocation points based on a minimum energy electrostatic problem was recently described in [16]. Both [1,16] propose and construct quality point distributions for triangular high order $h-p$ finite element methods. But for diagonal mass matrix spectral element methods we prefer Fekete points for the following reasons:

(1) In the square, with the diamond truncation, the Fekete points are the Gauss-Lobatto points [8], and thus a Fekete point spectral element method is the same as

the standard spectral element method. But unlike the Gauss-Lobatto points, Fekete points are defined for any domain and with any polynomial truncation.

(2) Theoretical [7] and numerical evidence suggests that Fekete points along the boundary of the triangle are the Gauss-Lobatto points, making Fekete point triangular and quadrilateral elements naturally conform. Other methods impose this as an additional constraint.

(3) The definition of Fekete points is closely related to the expression for cardinal functions (Lagrange interpolating polynomials). We believe this relation gives Fekete point cardinal functions which approximate delta functions and Fekete point quadrature weights which are all positive. Both of these are required properties for spectral element methods which have not been investigated in more than one dimension.

(4) The Fekete criterion leads to a simple algorithm which at present is able to generate more optimal interpolation points than other techniques (for large degree). Good interpolation is required for well behaved approximation, quadrature and derivative operators.

The only difficulty in using Fekete points is that they must be computed numerically. In the triangle, they have been approximated in [6] up to degree 7. These results were improved upon in [1] and extended to degree 13. Electrostatic based interpolation points were computed in [16], up to degree $d = 16$. In this work, we will present an algorithm for computing Fekete points along with results for the triangle for up to degree 19. In reality, we have no way of knowing if the algorithm computes true Fekete points. We simply use the Fekete criterion to compute points with all the properties outlined above. For degrees larger than 10, the results have the lowest Lebesgue constant of any previous points in the triangle.

We will first describe Fekete points in Sections 2 and 3. Then in Sections 4 and 5 the standard derivation of the spectral element method is outlined, this time using Fekete points in quadrilaterals or triangles. The algorithm to compute Fekete points is described in Section 6, and results are presented in Section 7.

2. An introduction to Fekete points. Fekete points are closely related to optimal interpolation points, so we first describe some well known facts about interpolation (see, for example [9]).

Let $\mathcal{P}_N$ be a finite dimensional vector space made up of polynomials, and let Ω be some domain such as the square or triangle. Setting $N = \dim \mathcal{P}_N$, let us now take N points, z_i, which we assume are solvable in $\mathcal{P}_N$. That is, given an arbitrary function f, there is a unique function $g \in \mathcal{P}_N$ where $g(z_i) = f(z_i)$. We will denote the interpolating polynomial g by

$$g(z) = I_N(f(z)).$$

One can then ask the question how well does $I(f)$ approximate f? To see this, take a function $h \in \mathcal{P}_N$ which best approximates f in the max norm,

$$\|f\| = \max_{z \in \Omega} |f(z)|.$$

This function is not necessarily $I_N(f)$, but $h = I_N(h)$. Thus,

$$\begin{aligned} \|f - I_N(f)\| &= \|f - h + I_N(h) - I_N(f)\| \\ &\leq \|f - h\| + \|I_N\| \, \|h - f\| \\ &\leq (1 + \|I_N\|) \, \|f - h\|, \end{aligned}$$

where the operator max-norm is defined in the usually way

$$\|I_N\| = \max_{\|f\|=1} \|I_N(f)\|.$$

If our functional space $\mathcal{P}_N$ is a good choice for approximating f, then $I_N(f)$ will be a good approximation to f if the norm of the interpolation operator $\|I_N\|$ is reasonable.

Choices such as equally spaced (in the square or triangle), or a product Gaussian grid (restricted to the triangle) lead to disasters. In these cases it is well known that the norm of the interpolation operator grows exponentially with the degree. If the points are not chosen very carefully, the interpolating polynomial will have wild oscillations between the collocation points (see for example, [9, p171]. This fundamental problem can not be overcome by after-the-fact preconditioning.

For interpolation we thus would like to find a set of points with small interpolation max-norm $\|I_N\|$. This norm is the Lebesgue constant, which is a function of the interpolation points, the domain Ω and the functional space $\mathcal{P}_N$. For a given Ω and $\mathcal{P}_N$, Lebesgue points are the points with minimum Lebesgue constant, and thus (in this norm) are the optimal interpolation points. Almost nothing seems to be known about Lebesgue points in more than one dimension. Nor are we aware of a feasible method for computing them numerically.

Fekete points are a more tractable alternative to Lebesgue points. There are some theoretical results for Fekete points in more than one dimension and numerical evidence shows they are close to optimal [1]. To define Fekete points for a domain Ω, we first pick a basis $\{g_i, i = 1..N\}$ for $\mathcal{P}_N$. Then let $V(z_1, z_2, ..., z_N)$ be the generalized Vandermonde matrix defined at the points $\{z_i \in \Omega, i = 1..N\}$. That is, V is an $N \times N$ matrix whose elements are $V_{ij} = g_j(z_i)$. Fekete points are a set of points $\{z_i, i = 1..N\}$ which maximize (for a fixed basis) the determinant of V:

$$\max_{\{z_i\}} |V(z_1, z_2, ..., z_N)|.$$

Fekete points are independent of our choice of basis, since any change of basis only multiples the determinant by a constant independent of the points. The matrix V can be thought of as the inverse transform matrix. V maps spectral coefficients of a function into that function's grid point values at z_i, and maximizing $|V|$ makes this matrix far from singular.

In the square, with $\mathcal{P}_N$ taken to be a diamond truncation of polynomials, Bos has recently proven that the Fekete points are unique and given by the tensor product of Gauss-Lobatto points [8]. This was proven for the interval $[-1, 1]$ in [12]. In one dimension, Fekete points are also the minimum energy configuration of point charges in the interval (attributed to Stieltjes in [27, p 140]). However, in higher dimensions, all of these similarities break down [5]. For general domains, Fekete points are not Gaussian-like quadrature points, nor will they be minimum energy electrostatic points (even though these latter points are also sometimes called Fekete points). For the electrostatic problem, the limiting distribution as $N \to \infty$ is known. For Fekete points, this limiting distribution is conjectured to be a certain extremal measure [5], but this is an unsolved problem. However, Hesthaven [16] has shown that a generalized form of the electrostatic problem with Gauss-Lobatto points imposed on the boundary will produces points with better Lebesgue constant than Fekete points for degree $d < 9$.

There are some analytic results for Fekete points in a triangle. If we assume the Vandermonde matrix is non-singular, then there is a maximum number of points which can lie on the boundary. Bos [7] conjectured that in the triangle (and some other domains) Fekete points will have the maximum number of points on the boundary. Under this assumption, he proved that the boundary points will be the one dimensional Gauss-Lobatto points. We have verified this conjecture numerically. This result has an important application: Fekete point triangular elements naturally conform with standard quadrilateral spectral elements.

3. Fekete point cardinal functions. As mentioned above, Fekete points corresponding to a finite dimensional polynomial truncation $\mathcal{P}_N$ and a domain Ω are the points which maximize the determinant of the generalized Vandermonde matrix $V(z_1, z_2, ..., z_N)$. Let us call this maximum v:

$$v = \max_{\{z_i\}\in\Omega} |V(\{z_i\})|.$$

We can now write an expression for the cardinal functions (Lagrange interpolation functions). If $v \neq 0$, the cardinal functions $\phi_j(z) \in \mathcal{P}_N$ are uniquely defined by

$$\phi_j(z_i) = \delta_{ij}$$

where δ_{ij} is the usual delta function ($\delta_{ij} = 1$ if $i = j$ and 0 otherwise). The cardinal functions defined at Fekete points can be written as

$$\phi_j(z) = \left.\frac{|V(\{z_i\})|}{v}\right|_{z_j=z}.$$

This is because at $z = z_i$ for $i \neq j$, two rows of the Vandermonde matrix are equal and $|V| = 0$. When $z = z_i$, we have that the determinant of V is at its maximum and thus $|V| = v$. This expression also leads to a bound on the cardinal functions

$$|\phi_i(z)| \leq 1, \qquad \forall z \in \Omega.$$

This bound allows us to bound the norm of the interpolation operator I_N. Writing $I_N(f)$ in terms of cardinal functions,

$$I_N(f) = \sum_{i=1}^{N} f(z_i)\phi_i(z)$$

we see that (in the max norm)

$$\begin{aligned} \|I_N\| &= \max_{\|f\|=1} \|I_N(f)\| \\ &= \max_{z\in\Omega} \sum_{i=1}^{N} |\phi_i(z)| \leq N. \end{aligned}$$

For the triangle, we will be using a triangular truncation of polynomials of degree d where $N = (d+1)(d+2)/2$. In the square, we use the standard diamond truncation of degree d where $N = (d+1)^2$.

This close relation between Fekete points and cardinal functions is important for spectral element methods. These methods rely on a accurately computing inner products of the form

$$\int_\Omega f\phi_i$$

by weighted quadrature at the collocation points [21]. Since the cardinal functions are discrete delta functions at these grid points, the quadrature will evaluate to $f(x_i)$ multiplied by the quadrature weight. The accuracy of the method depends on the accuracy of this quadrature approximation. This requires that the cardinal functions approximate (in the weak * sense) continuous weighted delta functions.

It the square, this convergence can be shown using the fact that the Fekete points are also Gauss-Lobatto quadrature points. For the triangle, this convergence is only conjectured. At present, all we can say is that unlike general optimal interpolation points, Fekete points generate cardinal functions which have the delta function quality of achieving their maximum (in Ω) at their associated Fekete point.

4. The spectral element discretization. We will now follow [21] and repeat the standard spectral element discretization, but for the square and the triangle using Fekete points.

Let us start with the model advection diffusion equation

$$u_t = -\mathbf{v} \cdot \nabla u + \epsilon \nabla^2 u.$$

The velocity field $\mathbf{v}$ is given, with no flow through the boundary. We will solve this equation for u, in a domain Ω, with the value of u at $t = 0$ given and for simplicity we impose Dirichlet boundary conditions for u. We use this equation because it contains terms similar to those in the primitive equations used it atmospheric and oceanic modeling. Furthermore, it can be solved without requiring the inversion of any elliptic operators, thus making the benefits of a diagonal mass matrix more obvious.

The first step is to write the weak or integral form of the equations. We multiply by a test function ψ and integrate over the entire domain Ω

$$\int_\Omega u_t \psi \, dA = -\int_\Omega \left(\psi \mathbf{v} \cdot \nabla u + \epsilon \nabla u \cdot \nabla \psi\right) \, dA,$$

where dA is the usual area measure and we have integrated the diffusion term by parts. The next step is to break the domain Ω into elements Ω_k,

$$\Omega = \bigcup_k \Omega_k.$$

We will be interested in the case were Ω_k can easilly be mapped to either a square or triangle.

The integral form of the advection diffusion equation can then be written

$$\sum_k \int u_t \psi J^k \, dA = -\sum_k \int \left(\mathbf{v} \cdot \nabla u \psi + \epsilon \nabla u \cdot \nabla \psi\right) J^k \, dA \tag{1}$$

integrated over the square (or triangle) and J^k is the Jacobian of the mapping from the square (or triangle) to Ω_k.

The spectral element descritization has three steps.

Step 1. The functional space. First we must choose a finite dimensional vector space of functions, $\mathcal{P}_N$. This space should well approximate smooth functions f. let $N = \dim \mathcal{P}_N$. For quadrilaterals, the best choice is the usual tensor product of polynomials of at most degree d (diamond truncation),

$$\mathcal{P}_N = \mathcal{P}^d(x) \times \mathcal{P}^d(y) = \operatorname{span}\{x^m y^n,\, m, n \leq d\},$$

where $\mathcal{P}^d(x)$ are the polynomials in one variable x of degree d. The dimension of this space is $N = (d+1)^2$.

The space used in triangles is slightly different. In this case $\mathcal{P}_N$ is almost always taken to be the standard triangular polynomial truncation

$$\mathcal{P}_N = \mathcal{P}^d(x,y) = \text{span}\{x^m y^n,\, m+n \le d\},$$

where $\mathcal{P}^d(x,y)$ are the polynomials in x and y of at most degree d. The dimension of this space is $N = (d+1)(d+2)/2$. An orthogonal basis with respect to dA and Gaussian type quadrature for this space was constructed by Dubiner[11]. The Dubiner basis for the right triangle, with base [0,1] and height [0,1] is given by

$$g_{mn} = P_m^{0,0}\left(\frac{x}{1-y}\right)(1-y)^m P_n^{2m+1,0}(y),$$

where $P_n^{\alpha,\beta}$ are the Jacobi Polynomials with weight (α,β) and degree n.

Further justification for this choice of functional space and choice of truncation is given in [31]. There it was discoverd that the Dubiner basis functions are also the eigenfunctions of the natural Sturm-Liouville problem for the triangle. The Dubiner basis g_{mn} has eigenvalue $(m+n)(m+n+2)$, so if one were to take for $\mathcal{P}_N$ the span of all eigenfunctions with eigenvalues less than a constant, then one again arrives at $\mathcal{P}^d(x,y)$. Finally, we note that this truncation of polynomials has the nice property that the space is invariant under all linear maps and thus invariant under all maps of one triangle into another. The truncation used with quadralaterals does not have this property.

Step 2. The collocation points. The choice of collocation points seems to be the most important step, since they govern the quality of the underlying interpolation operator which controls the accuracy of the numerical method.

We need to pick a set of collocation points $\{z_i = (x_i,y_i),\, i = 1..N\}$ for the square and triangle. We need exactly $N = \dim \mathcal{P}_N$ collocation points so that we can compute cardinal functions (Lagrange interpolating functions) $\phi_j \in \mathcal{P}_N$. The ϕ_j will then form our basis for $\mathcal{P}_N$. For correct coupling at the boundaries, we also require that the convex hull formed by the points is the boundary of the square or triangle, with (for a degree d truncation) exactly $d+1$ points along each edge of the square or triangle. This is why Gauss-Lobatto quadrature points are used in the square rather than Gauss quadrature points.

For quadrilaterals, the usual choice is a tensor product of the one dimensional Gauss-Lobatto quadrature points. In one dimension, these points are defined by the roots of

$$(1-x^2)\frac{dP^d}{dx}(x) = 0,$$

where $P^d(x)$ is the Legendre polynomial of degree d. A tensor product grid of the Gauss-Lobatto points has $N = (d+1)^2$ points, as required. As previously mentioned, these points are the Fekete points for the square with the diamond polynomial truncation $\mathcal{P}_N = \mathcal{P}^d(x) \times \mathcal{P}^d(y)$.

For the triangle, we also use Fekete points, but with the triangular polynomial truncation $\mathcal{P}_N = \mathcal{P}^d(x,y)$. Fortunately, these points also appear to be distributed as the degree d Gauss-Lobatto points along the edges of the triangle. Thus degree d

Fekete point square elements can be freely mixed with degree d Fekete point triangular elements without changing any of our derivation.

The choice of collocation points determines the cardinal functions which we use as basis functions. The expansion of our unknown variable u in this basis is defined by the interpolating polynomial $I_N(u)$,

$$I_N(u) = \sum_{i=1}^{N} u(z_i)\phi_i.$$

The choice of basis then determines our discrete derivative operators in the usual way. The gradients which appear in equation (1) are also computed in terms of I_N. For an arbitrary function f, we approximate ∇f by $\nabla I_N(f)$, where the later term is computed exactly by differentiating the basis functions ϕ_i exactly.

The global test functions ψ will also be constructed out of the ϕ_i. For each collocation point z_i^k (superscripted with a k to denote that it belongs to element Ω_k) we define a continuous and piecewise polynomial test function $\psi_i^k(z)$, $z \in \Omega$ as follows.

For $z_i^k \in$ the interior of Ω_k,

$$\psi_i^k(z) = \begin{cases} \phi_i(z) & z \in \Omega_k \\ 0 & z \notin \Omega_k \end{cases}$$

That is, we define the test function to be the cardinal function within Ω_k and zero outside Ω_k, creating a continuous and piecewise polynomial function.

For z_i^k on the boundary of Ω_k, then z_i^k may also be on the boundary of another element, or many elements if it is a corner point. This case is similar to above, except now we must piece the test function together out of the cardinal functions associated with all elements which contain z_i^k. For simplicity's sake, let us assume that there are only two such elements, which we label $k1$ and $k2$. The common boundary points shared by these elements we label $z_i^k = z_{i1} = z_{i2}$. The cardinal function belong to element $k1$ at the point z_{i1} is denoted ϕ_{i1} and similarly for ϕ_{i2}. Then we define our global test function ψ_i^k by

$$\psi_i^k(z) = \psi_{i1}^{k1}(z) = \psi_{i2}^{k2}(z) = \begin{cases} \phi_{i1}(z) & z \in \Omega_{k1} \\ \phi_{i2}(z) & z \in \Omega_{k2} \\ 0 & z \notin \Omega_{k1} \cup \Omega_{k2} \end{cases}$$

which is continuous and piecewise polynomial. This process constructs a unique global test function for each unique grid point.

Step 3. The quadrature. The final step is to replace the integrals (over the square or the triangle) in equation (1) by quadrature formula,

$$Q_N(f) = \sum_{i=1}^{N} w_i f(z_i). \tag{2}$$

Since the collocation points x_i are chosen above in step 2, our only freedom is the choice of quadrature weights w_i. These weights are chosen so that

$$Q_N(p) = \int p\, dA \quad \forall p \in \mathcal{P}_N.$$

This is achieved by taking

$$w_i = \int \phi_i \, dA. \tag{3}$$

Note that for the square, with the above weights and the tensor product Gauss-Lobatto points, the quadrature formula is more accurate than this construction implies. For quadrilaterals, we have that $Q_N(p)$ is exact not only for $p \in \mathcal{P}^d(x) \times \mathcal{P}^d(y)$, but actually for all $p \in \mathcal{P}^{2d-1}(x) \times \mathcal{P}^{2d-1}(y)$ [10]. However, this is still not accurate enough to integrate exactly all the terms that appear our linear advection diffusion equation (1). Even if we treat the velocity and the Jacobians as constants, some of the integrands in (1) could be of degree $2d$.

This last point is perhaps key to the simplicity of the spectral element method. If we forgo computing the integrals exactly, and instead replace them by approximate quadrature, we get a remarkable simplification: a diagonal mass matrix. Once this is done however, the placement of the collocation points becomes extremely important. They must be placed in such a way that the quadrature is very accurate. Bounds on the quadrature error for both the square and the triangle will be given in the next section.

The final discretization. We now generate a system of equations by evaluating (1) with each of the test functions ψ_i^k. We replace the integrals which appear in (1) by the quadrature formula Q_N and make use of the fact that the test functions are built out of discrete delta functions. For a test function ψ_i^k associated to an element interior collocation point $z = z_i^k$, the equations reduce to

$$w_i^k J^k \frac{\partial u}{\partial t} = -w_i^k J^k \mathbf{v} \cdot \nabla u + \epsilon Q_N \left(J^k \nabla u \cdot \nabla \psi_i^k \right),$$

evaluated at $z = z_i^k$. For a test function ψ_i^k associated with an element boundary point $z = z_i^k$, we will have the sum of terms coming from all other elements which contain z_i^k. Again we just treat the case where there are two such elements, labeled by $k1$ and $k2$. Then

$$\begin{aligned} \left(w_{i1}^{k1} J^{k1} + w_{i1}^{k1} J^{k2} \right) \frac{\partial u}{\partial t} = & \\ & -w_{i1}^{k1} J^{k1} \mathbf{v} \cdot \nabla u^{k2} + \epsilon Q_N \left(J^{k1} \nabla u^{k1} \cdot \nabla \psi_{i1}^{k1} \right) \\ & -w_{i2}^{k2} J^{k2} \mathbf{v} \cdot \nabla u^{k2} + \epsilon Q_N \left(J^{k2} \nabla u^{k2} \cdot \nabla \psi_{i2}^{k2} \right), \end{aligned}$$

evaluated at $z = z_i^k$.

It is now clear that the mass matrix is diagonal. With an explicit time stepping scheme used to evaluate the time derivatives, the final discretization is fully explicit.

Tensor product grids. There is one major advantage to using quadrilaterals that we have not yet emphasized: The tensor product grid. Let $n = d + 1$, so that within each quadrilateral there is an $n \times n$ grid of collocation points. Then many operations, such as differentiation, require n^2 sums to be computed over n^2 points. Because of the underlying product grid, this $O(n^4)$ operation can be computed in only $2n^3$ operations.

For triangular spectral element methods based on Fekete points, this is not the case. A derivative calculation requires $n^2(n+1)^2/2$ operations. For n large, this non-tensor product approach would be extremely expensive. But for values for n on the order of 8-16, this disadvantage is extremely overstated in the literature. Take for example $n = 16$, where computing a derivative on a Fekete point grid in the triangle requires 4.5 times more work then it would on a tensor product grid in the square. In practice, this difference will be almost negligible for several reasons: First, in a realistic model, the derivative and other $O(n^3)$ calculations represent just a fraction of the total work. Secondly, on modern cache based supercomputers, these small, dense computations fit entirely in cache and run at close to peak processor speed. The true performance bottleneck is the large, memory bandwidth limited time stepping loops which run at no better than 10% of peak. Finally, because of the conforming nature of Fekete point quadrilaterals and triangles, we could use mostly quadrilaterals and resort to triangles only near irregular boundaries or regions of changing resolution.

5. Accuracy of Fekete point spectral elements. At this point we have not said anything about the quality of the method. The fact that Fekete point cardinal functions are bounded by 1 will allow us to prove uniform and exponential convergence of interpolation, differentiation and quadrature when applied to an analytic function f. However, this does not seem to be enough to generalize the convergence proofs (for linear elliptic problems) given for the spectral element method in [21]. We quickly run into a problem which reduces to the fact that we do not have a good estimate as to how fast Fekete point cardinal functions converge to delta functions. This proof would require detailed knowledge of the structure of Fekete points, something which at present is only known in domains such as the square. For the triangle, we are optimistic that this will be true, but our only hope of showing it will be to build a numerical model and measure its convergence rate numerically.

In lieu of a convergence proof, we will give the standard error bounds for Fekete point approximation, differentiation and quadrature operators in a single domain. We will estimate all errors in terms of $\|f - h\|$ (or the first derivative max-norm, $\|f - h\|_{W_1^\infty}$), where $h \in \mathcal{P}_N$ is the best approximation to f. For an analytic function f, and $\mathcal{P}_N$ either the triangular or diamond polynomial truncations of degree d, as d is increased h will converge (in either norm) uniformly and exponentially fast to f.

The bound on approximation by Fekete point interpolation,

$$\|f - I_N(f)\| \leq (1 + \|I_N\|)\, \|f - h\|$$

was computed above. We have already seen that $\|I_N\| \leq N \leq Cd^2$. In the square, this bound can be improved to $\|I_N\| \leq C \log^2 d$. Numerical evidence presented in Section 7 indicates that in the triangle this bound is close to Cd. In any case, we see that for analytic f, the Fekete point polynomial approximation to f converge uniformly and exponentially fast to f.

The same derivation can be used to estimate the error in our discrete derivative operator $\nabla I_N(f)$

$$\begin{aligned}
\|\nabla f - \nabla I_N(f)\| &= \|\nabla f - \nabla h + \nabla I_N(h) - \nabla I_N(f)\| \\
&\leq \|\nabla f - \nabla h\| + \|\nabla I_N(f) - \nabla I_N(h)\| \\
&\leq (1 + Cd^2 \|I_N\|)\, \|f - h\|_{W_1^\infty}.
\end{aligned}$$

Here we have used the classic bound $\|\nabla\| < Cd^2$ when restricted to polynomials of degree d. Again we see that we can maintain exponential convergence to analytic functions if we have a reasonable bound on $\|I_N\|$.

The quadrature error estimate is similar. Since $Q_N(h)$ integrates $h \in \mathcal{P}_N$ exactly, we have

$$\begin{aligned}\|\int f\,dA - Q_N(f)\| &= \|\int f\,dA - \int h\,dA + Q_N(h) - Q_N(f)\| \\ &\le \|\int (f-h)\,dA\| + \|Q_N(f) - Q_N(h)\| \\ &\le (A + \|Q_N\|)\,\|f-h\|,\end{aligned}$$

where A is the area of the triangle or square. To estimate the norm of the quadrature operator, we refer back to equations (2) and (3), where it can be seen that the N weights are all bounded by the area A, implying $\|Q_N\| \le AN < Cd^2$. If the weights are positive, as they are for the square, (and numerical evidence suggests this is mostly true for Fekete points in the triangle), this bound can be improved to $\|Q_N\| < A$.

Thus for both the triangle and the square, for analytic f, the Fekete point approximation, differentiation and quadrature all converge uniformly and exponentially fast to the exact answer.

6. Computing Fekete points. We only know of a few works in which Fekete points are computed in more than one dimension. The earliest work is due to Bos [6], who derived Fekete points for the triangle up to degree $d = 3$, and derived some approximate solutions up to degree 7. More recent work has been computational. Chen and Babuška [1] improved and extended Bos's results to degree 13. They also computed optimal L^2-norm interpolation points, and showed that these points have a lower Lebesgue constant than their approximate Fekete points. Their algorithm is not described, but the minimization seems not to be effective for $d > 10$ since we will present Fekete points with smaller Lebesgue constant than both their Fekete and minimal L^2-norm points. Thus we suspect an improved algorithm could compute optimal L^2-norm points with even smaller Lebesgue constants.

Hesthaven [16] computed a different set of near-optimal interpolation points for the triangle. He uses a second order time evolution method to minimize an electrostatic energy function. The resulting solution is further refined with a modified Powell's hybrid method. For $d < 9$ these points are quite good, with a smaller Lebesgue constant than Fekete points and sometimes even the optimal L^2-norm points. This fails to be true for larger d, and for $d > 13$ these points become significantly worse than Fekete points. All of these results are summarized in Table I.

We note that Fekete points have also been computed on the sphere using the equal area resolution triangular truncation of spherical harmonics [23]. They use an exchange algorithm which is an iterative sequence of independent minimizations for each point.

6.1 Steepest ascent algorithm. The algorithm we use to maximize the determinant of the Vandermonde matrix is the steepest ascent method. We implement this with a simple gradient flow algorithm to solve the system of ODEs:

$$\frac{\partial z_i}{\partial t} = \frac{\partial |V|}{\partial z_i}.$$

The points are evolved by moving them in the direction of steepest accent until an equilibrium solution is reached. The only constraint is that the points are not allowed to leave the triangle. We perform these computations in the right triangle. The algorithm will terminate at a local maximum of $|V|$, and we call any such extremal point a solution. In practice we can find several different solutions and we evaluate these solutions with a variety of norms. At present we have know way of knowing if any of our computed solutions represent a global maximum and thus represent the true Fekete points, hence we refer to these local maximum solutions as approximate Fekete points. Furthermore, faced with several sets of approximate Fekete points, we do not take the set with the largest value for $|V|$, but rather the set with positive quadrature and smallest Lebesgue constant.

The algorithm requires us to compute $\nabla|V|$, which has a simple expression if we write V in terms of a cardinal function basis. To show this, we first note that the partial derivative of the a determinant of a matrix with respect to an entry V_{ij} is given by

$$\frac{\partial|V|}{\partial V_{ij}} = -1^{i+j}|A_{ij}|$$

where A_{ij} is the ij minor of V (the matrix formed by removing the i'th and j'th rows of V). When V is computed using a cardinal function basis, V is the Identity matrix. Thus the derivative of $|V|$ with respect to any matrix element is only non-zero for elements along the diagonal. When V is written with a cardinal function basis, the i'th diagonal element is the i'th cardinal function evaluated at the grid points z_i and the determinant of the minor $|A_{ii}| = 1$. Thus our system of ODEs reduces to

$$\frac{\partial z_i}{\partial t} = \frac{\partial \phi_i}{\partial z_i}$$

This algorithm has a very simple geometric interpretation. We would like the cardinal functions ϕ_i to look like delta functions at z_i. The maximum should be achieved at the grid point z_i. The steepest ascent algorithm simply movies each point towards the maximum of its associated cardinal function. The iterative nature of the algorithm comes into play because the cardinal functions change with every change in the grid points, and thus we must recompute our basis functions at each iteration. The iteration is terminated when

$$\left|\frac{\partial \phi_i}{\partial z_i}\right| \leq 10^{-12}$$

for all z_i in the interior of the triangle.

The only remaining computational issue is in evaluating the cardinal functions and their derivatives. This is done by first computing the spectral coefficients of the cardinal functions with respect to a known basis. If the Vandermonde matrix is written using this known basis, then the coefficients of the expansion of the cardinal functions in terms of this basis are given by the rows of V^{-1}. Thus we must invert V (through Gaussian elimination) at each iteration. Since we must compute this inverse numerically, it is important that V be well conditioned. For this we have found that the Dubiner basis functions in the right triangle are far superior to Legendre polynomials. Using monomials for the basis functions is practically impossible. For our largest case, degree 19, the Matlab reported condition number of V is less than 50 when computed with the Dubiner polynomials.

A potential drawback of the steepest descent method is its slow rate of convergence. At present, a degree 19 maximization running in Matlab on a Sun workstation takes only a few hours to converge to 12 digits of accuracy. To compute Fekete points for larger degree, we will need a more sophisticated maximization algorithm and implementation.

6.2 Initial Conditions. The method is extremely sensitive to the initial condition for the grid points. With different initial conditions, we can find different local maximums of $|V|$. This makes it important to initialize the algorithm with a good initial guess. We have tried several possibilities, the best of which turns out to be points distributed to generate a density which approximates the extremal measure $g(x,y)$ for the triangle given in [2]. For the right triangle $x \geq 0$, $y \geq 0$ and $x+y \leq 1$, the extremal measure is

$$g(x,y) = \frac{1}{\sqrt{xy(1-x-y)}}.$$

We are motivated to use this extremal measure because it is conjectured that $g(x,y) = f(x,y)$, where $f(x,y)$ is the limiting distribution of the Fekete points (as the degree approaches infinity) [5]. The limit $f(x,y)$ is known to exist and to satisfy [18]

$$g(x,y) \leq Cf(x,y).$$

To distribute a finite set of points to approximate a given density $g(x,y)$ is not so easy. Our initial approximation, which has much room for improvement, is as follows. We first assume the points, at least topologically, form a nested family of triangles. Thus we compute a nested family of triangular shells which have a "mass" proportional to the number of points we have decided to place in that triangular shell. If there are k points to be placed in a given shell, we then chop that shell into k pieces, all with the same mass, and place one point in the center of each piece. This procedure produces points which respect the D_3 symmetry of the triangle, and the iterative method maintains this symmetry. Points in the triangle with such symmetry have orbits of either 1, 3 or 6 points. For a given set of points, there will be a variety of symmetry configurations which can be generated by altering the number of points within each shell, the number of shells, and the distribution of the points within a shell.

The algorithm does not place points along the edge of the triangle, although it does place a total of $3d$ points in the outermost triangular shell. In all cases we have computed, the iterative scheme quickly moves these points to the boundary of the triangle, and then aligns them to be the Gauss-Lobatto points, as conjectured. This remains true even when the initial distribution is not given D_3 symmetry. For $d > 7$, asymmetric initial data can lead to asymmetric solutions, but the boundary points are still the Gauss-Lobatto points.

The algorithm has also been used to compute Fekete points in the square, with slight modifications. For the square we the usual diamond polynomial truncation. Our initialization procedure uses a nested family of square shells and the extremal measure for the square. This initial condition retains the D_4 symmetries of the square, but is not a product grid. In all cases the algorithm converges to the exact solution: a tensor product of Gauss-Lobatto points.

7. Fekete points in the triangle. We now present our computed approximate Fekete points. We use three metrics to evaluate the quality of these and other sets of points in the triangle. The first metric is the Lebesgue constant, $\|I_N\|$. The cardinal

Table I

Lebesgue constants for point distributions in the triangle using the triangular polynomial truncation (unless noted otherwise). Key: "Fekete": approximate Fekete points from this paper. "CB Fekete": Chen and Babuška approximate Fekete points. "CB L^2": Chen and Babuška optimal L^2-norm points. "H" Hesthaven electrostatic points. "GL" Fekete points (Gauss-Lobatto points) in the square with the diamond polynomial truncation. For the "Fekete" points only, †denotes the presence of negative quadrature weights and ‡denotes an asymmetric solution.

degree	Fekete	CB Fekete	CB L^2	H	GL
6	4.17	4.17	3.79	4.07	3.50
7	4.91	4.93	4.39	4.78	3.89
8	9.43, 5.90†	5.90	5.09	5.88	4.18
9	6.80	6.80	5.92	6.92	4.50
10	8.11, 7.75†	8.00	7.09	8.40	4.71
11	8.76, 7.89†	9.52	8.34	10.09	5.02
12	9.60, 8.03†	11.08	10.08	12.52	5.20
13	9.21	13.24	12.05	15.34	5.49
14	10.8, 9.72†			22.18	5.62
15	9.97			29.69	5.91
16	12.1			41.73	6.08
17	13.3†				6.29
18	13.5				6.41
19	14.2‡				6.63

functions needed to compute the Lebesgue constant are computed as described in Section 6.1, and the maximum over all points in the triangle is approximated by the maximum over a grid of 2485 equally spaced points. These numbers are given for several different point distribution in Table I. For comparison, we also list the Lebesgue constant for the Fekete points (Gauss-Lobatto points) in the square. Note that the actual Lebesgue constant $\|I_N\|$ is better than the estimate given in Section 5. The values for the triangle can be seen to grow at a rate close to d. In the square, the Lebesgue constant is significantly better, suggesting that triangular spectral element methods will never be as accurate as quadrilateral methods.

Our second metric is whether or not all the quadrature weights for a given point distribution are positive. These weights are computed by solving the linear system

$$\sum_{j=1}^{N} w_j g_i(z_j) = \int g_i \, dA$$

for a basis $\{g_i, i = 1..N\}$ of $\mathcal{P}_N$. This requires the inverting the Vandermonde matrix, so again it is important that this matrix be well conditioned. Negative weights call into question the accuracy and stability of the associated quadrature and would be unusable in a spectral element method. Referring back to equation (3), a quadrature weight for a given point is given by the integral of its associated cardinal function, so a negative weight is an indication that the cardinal function is not a very good approximation of a delta function.

For spectral element methods, with their strong reliance on accurate quadrature, we believe a positive quadrature grid is preferable over other grids with slightly better Lebesgue constants. Thus in Table I, we list two approximate Fekete grids for a given degree if the grid with the lowest Lebesgue constant is non-positive. In all cases, when negative weights are present, they are limited to points on the edge of the triangle. One could modify the weights to make them positive, but any change

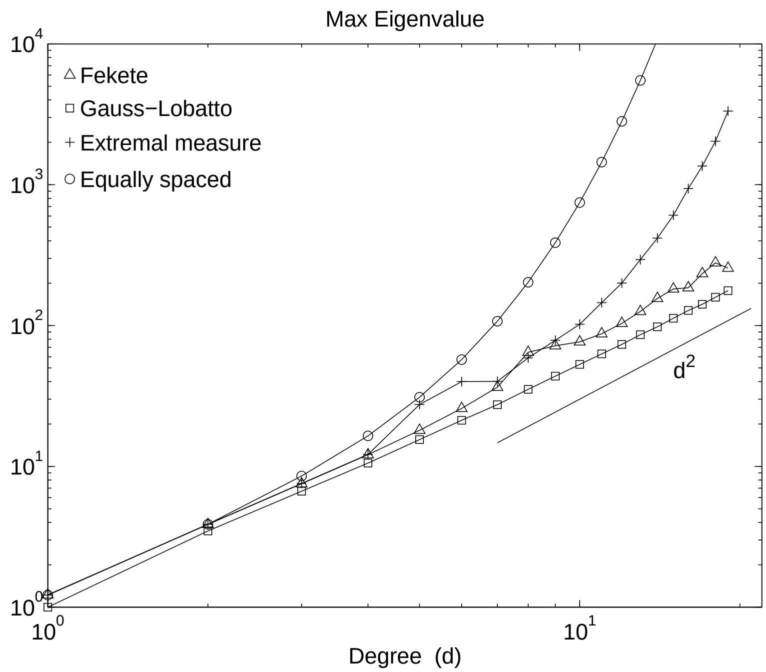

FIGURE 1. *Largest eigenvalue of the discrete derivative operator. Results are for the triangle with a triangular polynomial truncation, except the Gauss-Lobatto points, which are defined in the square using a diamond polynomial truncation.*

in the quadrature formula would mean it no longer integrates all of $\mathcal{P}_N$ exactly. The quadrature properties for the other interpolation points in Table I have not yet been investigated.

The third metric is the absolute value of the largest eigenvalue $|\lambda_{\max}|$ of the discrete derivative operator $\nabla I_N(f)$. This eigenvalue is closely related to $\|I_N\|$ since large oscillations between grid points requires a large derivative at the grid points. It also directly controls the CFL restriction on the time step for initial value problems. Like the Lebesgue constant, $|\lambda_{\max}|$ depends on the points, polynomial truncation and the domain but not on the choice of basis functions used to compute it. However, to compute $|\lambda_{\max}|$ accurately, one is almost required to use an orthogonal basis in the triangle such as the Dubiner polynomials. Computing $|\lambda_{\max}|$ is easier than computing $\|I_N\|$ since it does not involve a maximum over the whole triangle. In Figure 1, we plot $|\lambda_{\max}|$ for the Fekete points, equally spaced points and one family of extremal measure initial condition points in the triangle with the triangular polynomial trunctation. For comparison, we also plot the Fekete points (Gauss-Lobatto points) in the square with the diamond truncation. For Fekete points in both triangles and squares, this quantity is proportional to d^2. Thus (as in conventional spectral element methods) doubling the spectral degree results in a time step four times smaller.

Finally we present several computed approximate Fekete points in Figures 2 and 3. Next to each point distribution is a plot of the cardinal function associated with the leftmost corner point. This cardinal function is chosen because it is associated with the smallest (and sometimes negative as indicated in Table I) quadrature weight. The cardinal function values are plotted along a line from the corner point through the center of the triangle. Experience has shown that the largest amplitude oscillations of the cardinal function will be along this line. The grids plotted in Figure 2 all have positive quadrature and the cardinal function do appear to be converging (in the weak * norm) to a delta function on the triangle. The plots in Figure 3 have non-positive quadrature and the cardinal functions are more erratic.

8. Conclusions. We have outlined a new spectral element method based on Fekete points, and presented an algorithm to compute approximate Fekete points in the triangle. This algorithm has room for improvement, but has calculated grids with small Lebesgue constant and positive quadrature for up to degree 19. Fekete points for a dimension N polynomial space have N grid points, thus allowing the construction of cardinal functions at the Fekete points. Also, the Fekete criterion leads to near optimal grids with positive quadrature weights. These two properties allow us to construct a spectral element method with a diagonal mass matrix.

Our numerical computation of Fekete points verify a conjecture due to Bos [7] that the boundary Fekete points will be the one dimensional Gauss-Lobatto points. This allows coupling of Fekete point triangular spectral methods with the standard quadrilateral spectral element method. This later method is also a Fekete point method, since the tensor product of Gauss-Lobatto points are the Fekete points for the square.

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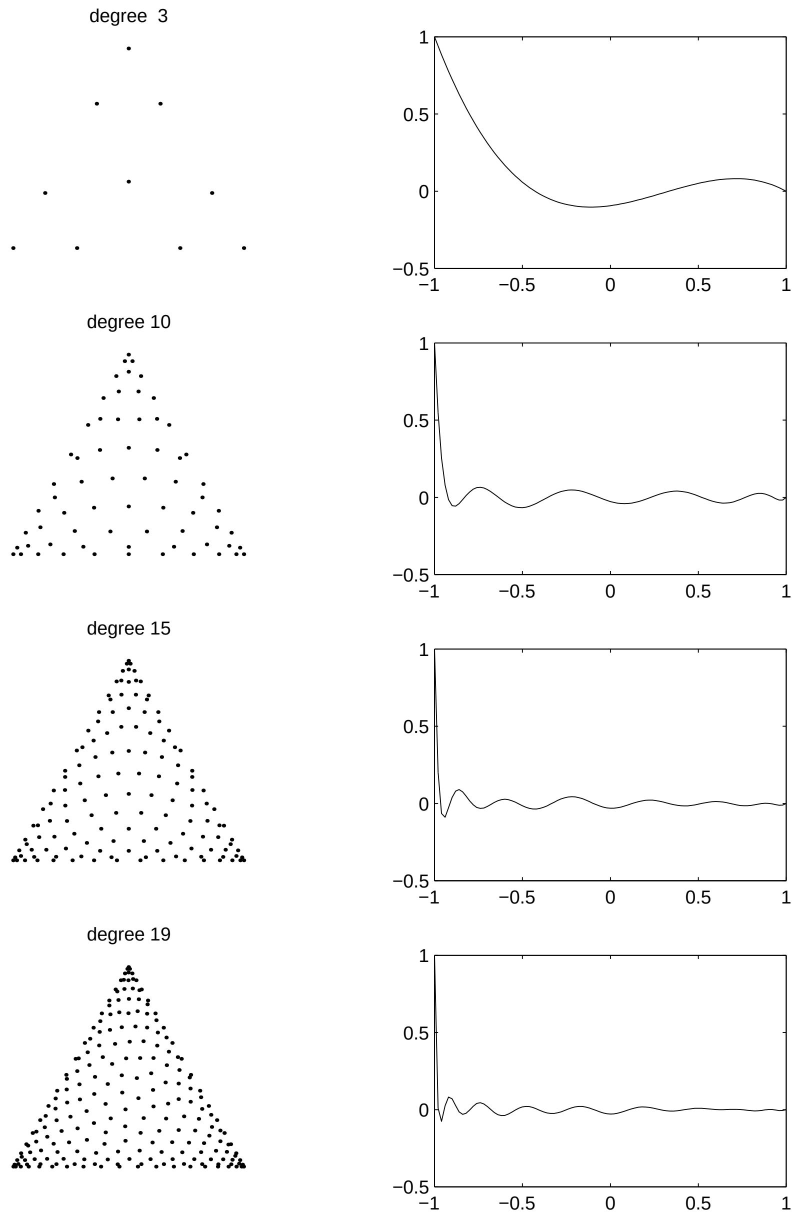

FIGURE 2. *Approximate Fekete grids with positive quadrature. Adjacent to each grid is the cardinal function associated with the left most corner point, plotted along the line through the center of the triangle.*

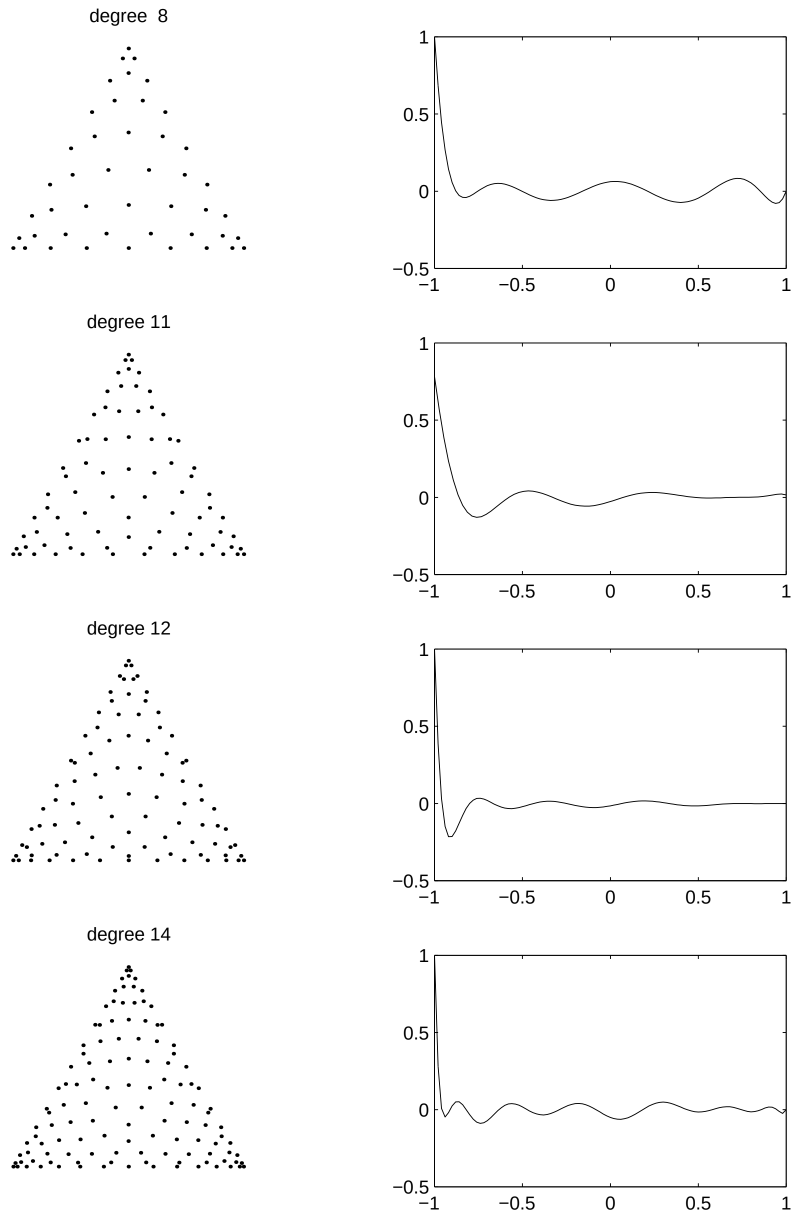

FIGURE 3. *Approximate Fekete grids with non-positive quadrature. Adjacent to each grid is the cardinal function associated with the left most corner point, plotted along the line through the center of the triangle.*

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Appendix I. Tables of approximate Fekete points. We now give the coordinates for selected approximate Fekete points. We list the grids which by symmetry must have a point in the center of the triangle, since it can be seen in Table I that these grids tend to have the best Lebesgue constants. For degree 12, we give both the positive and non-positive quadrature grids mentioned in Table I. Coordinates for all grids are available via email from the first author. We list the first two barycentric coordinates of each point (equivalent to the x and y coordinates after an equilateral triangle is linearly mapped to the right triangle with unit length base and height) along with the quadrature weight `w`. The third barycentric coordinate is defined such that the sum of all three coordinates is one. Only one point is listed from points possessing symmetries of the triangle. Such a point represents `orbit`=1,3 or 6 total points via permutations of its three barycentric coordinates.

```
d= 3:
orbit=1    0.3333333333   0.3333333333  w= 0.9000000000
orbit=3    0.0000000000   0.0000000000  w= 0.0333333333
orbit=6    0.0000000000   0.2763932023  w= 0.1666666667
d= 6:
orbit=1    0.3333333333   0.3333333333  w= 0.2178563571
```

```
orbit=3    0.1063354684   0.1063354684   w= 0.1104193374
orbit=3    0.0000000000   0.5000000002   w= 0.0358939762
orbit=3    0.0000000000   0.0000000000   w= 0.0004021278
orbit=6    0.1171809171   0.3162697959   w= 0.1771348660
orbit=6    0.0000000000   0.2655651402   w= 0.0272344079
orbit=6    0.0000000000   0.0848854223   w= 0.0192969460
d= 9:
orbit=1    0.3333333333   0.3333333333   w= 0.1096011288
orbit=3    0.1704318201   0.1704318201   w= 0.0767491008
orbit=3    0.0600824712   0.4699587644   w= 0.0646677819
orbit=3    0.0489345696   0.0489345696   w= 0.0276211659
orbit=3    0.0000000000   0.0000000000   w= 0.0013925011
orbit=6    0.1784337588   0.3252434900   w= 0.0933486453
orbit=6    0.0588564879   0.3010242110   w= 0.0619010169
orbit=6    0.0551758079   0.1543901944   w= 0.0437466450
orbit=6    0.0000000000   0.4173602935   w= 0.0114553907
orbit=6    0.0000000000   0.2610371960   w= 0.0093115568
orbit=6    0.0000000000   0.1306129092   w= 0.0078421987
orbit=6    0.0000000000   0.0402330070   w= 0.0022457501
d=12:
orbit=1    0.3333333333   0.3333333333   w= 0.0626245179
orbit=3    0.1988883477   0.4005558262   w= 0.0571359417
orbit=3    0.2618405201   0.2618405201   w= 0.0545982307
orbit=3    0.0807386775   0.0807386775   w= 0.0172630326
orbit=3    0.0336975736   0.0336975736   w= 0.0142519606
orbit=3    0.0000000000   0.5000000000   w= 0.0030868485
orbit=3    0.0000000000   0.0000000000   w= 0.0004270742
orbit=6    0.1089969290   0.3837518758   w= 0.0455876390
orbit=6    0.1590834479   0.2454317980   w= 0.0496701966
orbit=6    0.0887037176   0.1697134458   w= 0.0387998322
orbit=6    0.0302317829   0.4071849276   w= 0.0335323983
orbit=6    0.0748751152   0.2874821712   w= 0.0268431561
orbit=6    0.0250122615   0.2489279690   w= 0.0237377452
orbit=6    0.0262645218   0.1206826354   w= 0.0177255972
orbit=6    0.0000000000   0.3753565349   w= 0.0043097313
orbit=6    0.0000000000   0.2585450895   w= 0.0028258057
orbit=6    0.0000000000   0.1569057655   w= 0.0030994935
orbit=6    0.0000000000   0.0768262177   w= 0.0023829062
orbit=6    0.0000000000   0.0233450767   w= 0.0009998683
d=12:    (some negative quadrature weights)
orbit=1    0.3333333333   0.3333333333   w= 0.0485965670
orbit=3    0.2201371125   0.3169406831   w= 0.0602711576
orbit=3    0.2201371125   0.4629222044   w= 0.0602711576
orbit=3    0.1877171129   0.1877171129   w= 0.0476929767
orbit=3    0.1403402144   0.4298298928   w= 0.0453940802
orbit=3    0.0833252778   0.0833252778   w= 0.0258019417
orbit=3    0.0664674598   0.0252297247   w= 0.0122004614
```

```
orbit=3    0.0218884020   0.4890557990  w= 0.0230003812
orbit=3    0.0252297247   0.0664674598  w= 0.0122004614
orbit=3    0.0000000000   0.5000000000  w= 0.0018106475
orbit=3    0.0000000000   0.0000000000  w=-0.0006601747
orbit=6    0.1157463404   0.2842319093  w= 0.0455413513
orbit=6    0.0672850606   0.3971764400  w= 0.0334182802
orbit=6    0.0909839531   0.1779000668  w= 0.0324896773
orbit=6    0.0318311633   0.3025963402  w= 0.0299402736
orbit=6    0.0273518579   0.1733665506  w= 0.0233477738
orbit=6    0.0000000000   0.3753565349  w= 0.0065962854
orbit=6    0.0000000000   0.2585450895  w= 0.0021485117
orbit=6    0.0000000000   0.1569057655  w= 0.0034785755
orbit=6    0.0000000000   0.0768262177  w= 0.0013990566
orbit=6    0.0000000000   0.0233450767  w= 0.0028825748
d=15:
orbit=1    0.3333333333   0.3333333333  w= 0.0459710878
orbit=3    0.2379370518   0.3270403780  w= 0.0346650571
orbit=3    0.3270403780   0.2379370518  w= 0.0346650571
orbit=3    0.1586078048   0.4206960976  w= 0.0384470625
orbit=3    0.2260541354   0.2260541354  w= 0.0386013566
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orbit=3    0.0477095725   0.4761452137  w= 0.0243531004
orbit=3    0.0531173538   0.0531173538  w= 0.0094392654
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orbit=6    0.1585345951   0.3013819154  w= 0.0305412307
orbit=6    0.0972525649   0.3853507643  w= 0.0262101254
orbit=6    0.0875150140   0.2749910734  w= 0.0265367617
orbit=6    0.1339547708   0.1975591066  w= 0.0269859772
orbit=6    0.0475622627   0.3524012205  w= 0.0172635676
orbit=6    0.0596194677   0.1978887556  w= 0.0188795851
orbit=6    0.0534939782   0.1162464503  w= 0.0158224870
orbit=6    0.0157189888   0.4176001732  w= 0.0127170850
orbit=6    0.0196887324   0.2844332752  w= 0.0164489660
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orbit=6    0.0171941515   0.0816639421  w= 0.0072268907
orbit=6    0.0000000000   0.4493368632  w= 0.0023599161
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orbit=6    0.0000000000   0.0503997335  w= 0.0009919379
orbit=6    0.0000000000   0.0152159769  w= 0.0004893506
d=18:
orbit=1    0.3333333333   0.3333333333  w= 0.0326079297
orbit=3    0.2515553103   0.3292984162  w= 0.0255331366
orbit=3    0.3292984162   0.2515553103  w= 0.0255331366
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orbit=3	0.2438647767	0.2438647767	w= 0.0279490452
orbit=3	0.1512564554	0.1512564554	w= 0.0174438045
orbit=3	0.0810689493	0.4594655253	w= 0.0203594338
orbit=3	0.0832757649	0.0832757649	w= 0.0113349170
orbit=3	0.0369065587	0.0369065587	w= 0.0046614185
orbit=3	0.0149574850	0.0149574850	w= 0.0030346239
orbit=3	0.0000000000	0.5000000000	w= 0.0012508731
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orbit=6	0.1821465920	0.3095465041	w= 0.0235716330
orbit=6	0.1246901255	0.3789288931	w= 0.0206304700
orbit=6	0.1179441386	0.2868915642	w= 0.0204028340
orbit=6	0.1639418454	0.2204868669	w= 0.0215105697
orbit=6	0.0742549663	0.3532533654	w= 0.0183482070
orbit=6	0.0937816771	0.2191980979	w= 0.0174161032
orbit=6	0.0890951387	0.1446273457	w= 0.0155972434
orbit=6	0.0409065243	0.4360543636	w= 0.0119269616
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orbit=6	0.0132313129	0.2968103667	w= 0.0086477936
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orbit=6	0.0000000000	0.4154069883	w= 0.0006513246
orbit=6	0.0000000000	0.3332475761	w= 0.0021137942
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